








# Advanced Time Series Analysis and Forecasting

## Seminar 1: Forecast evaluation, scoring rules and combination




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# Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARX**: AutoRegressive model with eXogenous variables
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CDF**: Cumulative Distribution Function
- **COVID**: COronaVIrus Disease (2019 pandemic)
- **CRPS**: Continuous Ranked Probability Score
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **ES**: Expected Shortfall
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

# Acronyms used in this material (2/2)

- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **IS**: In-Sample
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MCS**: Model Confidence Set
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **PIT**: Probability Integral Transform
- **RMSE**: Root Mean Squared Error
- **RTDSM**: Real-Time Data Set for Macroeconomists (Philadelphia Fed)
- **SCA**: Seasonally and Calendar Adjusted
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **SPF**: Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia)
- **US**: United States

## Today's question and seminar structure

- The question of the seminar
  - **how do we decide, with the right inference, that one forecast is better than another, and when does averaging forecasts help?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

# Exercise map

| Task | Question | Type | Model |
|---|---|---|---|
| A1, A2 | optimal point forecasts: squared, absolute, pinball, LinEx | Solved, Proposed | A1 |
| A3, A4 | log score and CRPS; an improper score; the interval score | Solved, Proposed | A3 |
| A5, A6 | DM and HLN by hand; Clark–West by hand | Solved, Proposed | A5 |
| A7, A8 | Bates–Granger weights; the cost of estimated weights | Solved, Proposed | A7 |
| B1, B2 | Romanian GDP growth; combining SPF unemployment forecasts | Solved, Proposed | B1 |
| B3–B5 | EUR/RON densities; GDP vintages; load quantiles at the evening peak | Solved, Proposed | B1, B3 |
| C1, C2 | inflation forecasts by horizon; what is wrong in an AI answer? | Proposed | B1, A1–A8 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; its solution is discussed in the seminar
- **A**: calculations and derivations on paper; **B**: real data, solved in the notebook, each closing with an interpretation question; **C**: open questions and the critique of an AI answer

# Data used

| Series | Source | Frequency | Period |
|---|---|---|---|
| real GDP, Romania (SCA) | Eurostat (namq_10_gdp) | quarterly | 1995–2026 |
| SPF individual forecasts | Philadelphia Fed | quarterly | 1990–2025 |
| unemployment rate, US | FRED (UNRATE) | monthly | 1990–2026 |
| EUR/RON | BNR reference rate | daily | 2010–2026 |
| real GDP growth, all releases, US | Philadelphia Fed (RTDSM) | quarterly | 1990–2025 |
| electricity load, Romania | Energy-Charts (ENTSO-E data) | hourly | 2023–2026 |
| HICP inflation, Romania | Eurostat (prc_hicp_minr) | monthly | 1997–2026 |

- In the notebook: `dm_test`, `cw_test`, `mz_test`, `berkowitz_test`, `crps_normal`, `combine_spf`, `load_forecasts`; no account or key is needed

# Prerequisites for Today (1/24): forecasts and forecast errors

- Details: Chapter 1
- At the **origin** $t$ we forecast the **target** $y_{t+h}$, $h$ periods ahead (the **horizon**)

$$\hat{y}_{t+h|t} = \text{a function of } \mathcal{F}_t, \qquad e_{t+h} = y_{t+h} - \hat{y}_{t+h|t}$$

  - $\mathcal{F}_t$: the information available at $t$; $e_{t+h}$: the **forecast error**
  - **recursive** (expanding) window: estimate on all data up to $t$; **rolling** window: on the last $w$ observations only
- Benchmarks and simple models
  - historical mean $\hat{y}_{t+h|t} = \bar{y}_t$; **no change** (random walk) $\hat{y}_{t+h|t} = y_t$
  - AR(1): $\hat{y}_{t+\mathbf{1}|t} = \hat{c} + \hat{\phi} y_t$; **direct** AR($p$) at horizon $h$: regress $y_{t+h}$ on $y_t, \ldots, y_{t-p+\mathbf{1}}$

## Prerequisites for Today (2/24): accuracy measures

- Over $P$ out-of-sample forecasts with errors $e_1, \ldots, e_P$

$$\mathrm{RMSE} = \sqrt{\frac{1}{P}\sum_{t=1}^{P} e_t^2}, \qquad \mathrm{MAE} = \frac{1}{P}\sum_{t=1}^{P} |e_t|$$

  - RMSE (root mean squared error) and MAE (mean absolute error): in the units of $y$; lower is better
  - RMSE weights large errors more; MSE $= \mathrm{RMSE}^2$
- **Relative RMSE**: $\mathrm{RMSE}_{\text{model}}/\mathrm{RMSE}_{\text{benchmark}}$
  - below 1: the model beats the benchmark in the sample; whether the gap is significant needs a test (Diebold–Mariano, below)
- A **loss function** $L(x, y)$ prices a forecast $x$ when $y$ occurs
  - most losses depend only on the error $e = y - x$: $L(e)$

# Prerequisites for Today (3/24): loss and the optimal forecast

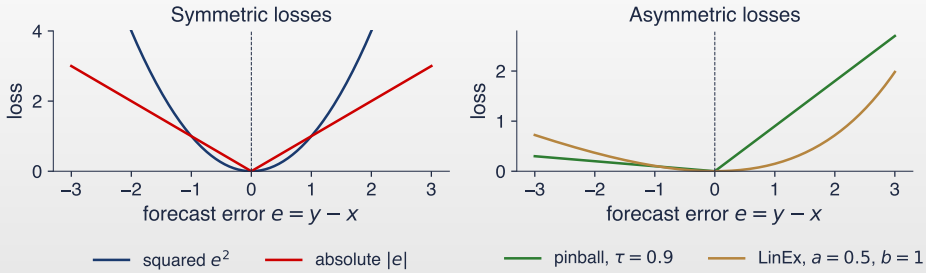
- $Y \sim F$: the **predictive distribution** of the target; the **optimal** forecast minimises the expected loss

$$x^* = \arg\min_x \mathbb{E}_F L(x, Y)$$

  - the minimum, $\mathbb{E}_F L(x^*, Y)$, is the **risk** of the optimal forecast
  - $L$ is **consistent** for a functional (mean, quantile) if that functional is optimal for every $F$

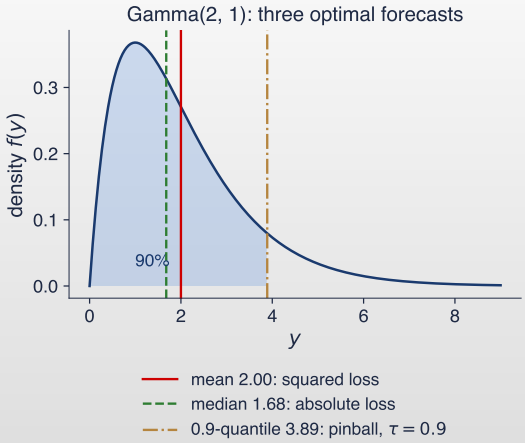
- Three losses and their optimal forecasts
  - squared $e^{\mathbf{2}} \to$ the mean $\mathbb{E}Y$; absolute $|e| \to$ the median $F^{-\mathbf{1}}(1/2)$
  - **pinball** $\rho_\tau(e) = (\tau - 1\{e < 0\})\, e \to$ the $\tau$-quantile $F^{-\mathbf{1}}(\tau)$
  - $\tau \in (0, 1)$: the quantile level; $1\{e < 0\} = 1$ if the forecast is too high, 0 otherwise
  - first-order condition of the pinball loss: $F(x) - \tau = 0$

# Prerequisites for Today (4/24): symmetric and asymmetric losses




- Left: squared loss punishes large errors much more than the absolute loss
- Right: with $\tau = 0.9$ an under-forecast ($e > 0$) costs 0.9 per unit, an over-forecast only 0.1; LinEx is exponential on one side

# Prerequisites for Today (5/24): optimal forecasts under three loss functions




- A right-skewed predictive distribution: Gamma(2, 1)
- Each loss asks for a different number
  - squared: the mean
  - absolute: the median
  - pinball with $\tau = 0.9$: the point with 90% of the probability to its left
- A forecaster must know the loss **before** forecasting
- Exponential($\lambda$): $F(y) = 1 - e^{-\lambda y}$, $y \geq 0$, mean $1/\lambda$

# Prerequisites for Today (6/24): the LinEx loss

- **LinEx**: linear on one side, exponential on the other

$$L(e) = b\,[e^{ae} - ae - 1], \qquad b > 0$$

  - $b$: the scale; $a$: the asymmetry; $a > 0$: under-forecasts ($e > 0$) cost exponentially, over-forecasts about linearly
  - $a \to 0$: $L(e) \approx ba^2e^2/2$, the squared loss

- Optimal forecast

$$x^* = \frac{1}{a} \ln \mathbb{E}e^{aY}$$

  - $\mathbb{E}e^{aY}$: the moment generating function of $Y$ at $a$

- For $Y \sim N(\mu, \sigma^2)$

$$\mathbb{E}e^{aY} = \exp(a\mu + a^2\sigma^2/2)$$

  - $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$: the target is Normal given the information at the origin; $\mu_t$, $\sigma_t^2$: conditional mean and variance

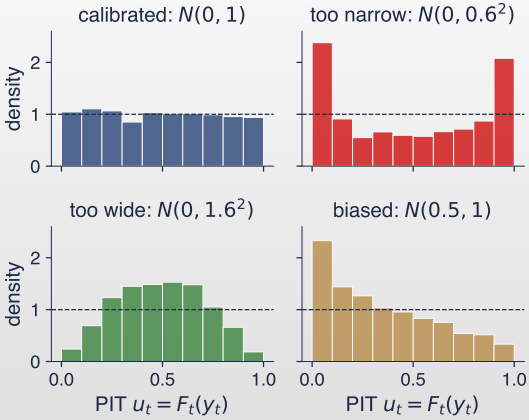
# Prerequisites for Today (7/24): density forecasts and the PIT

- A **density forecast** gives the whole predictive distribution
  - $F_t$: the forecast distribution function of $y_{t+\mathbf{1}}$; $f_t$: its density
- **PIT** (probability integral transform): the forecast probability of an outcome below the realised one

$$u_t = F_t(y_{t+\mathbf{1}}) \in [0, 1]$$

  - if $F_t$ is the true one-step conditional distribution, the $u_t$ are i.i.d. $U(0, 1)$ Diebold, Gunther and Tay (1998)
  - $U(0, 1)$: the uniform distribution on $[0, 1]$; its histogram is flat
- **Calibration** checks
  - share of $u_t$ outside $[0.05, 0.95]$: about 10% for a calibrated forecast
  - **sharpness**: how concentrated $F_t$ is; the goal is maximal sharpness subject to calibration

# Prerequisites for Today (8/24): reading a PIT histogram




- 2000 outcomes from $N(0, 1)$, four forecasts
- Shapes
  - flat: calibrated
  - U shape: too many outcomes in both tails, the forecast is **too narrow**
  - hump: **too wide**
  - slope: **biased**; here the forecast mean is too high, so small $u_t$ are too frequent
- Dashed line: the flat density 1 of $U(0, 1)$

# Prerequisites for Today (9/24): the Berkowitz test

- **Berkowitz** Berkowitz (2001): map the PITs to the Normal scale

$$z_t = \Phi^{-1}(u_t), \qquad z_t - \mu = \rho(z_{t-1} - \mu) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

  - $\Phi^{-1}$: the quantile function of $N(0,1)$; under correct forecasts $z_t$ i.i.d. $N(0,1)$, i.e. $\mu = 0$, $\sigma = 1$, $\rho = 0$

- **Likelihood ratio** (LR) statistic

$$\mathrm{LR} = 2\left[\ell(\hat{\mu}, \hat{\sigma}, \hat{\rho}) - \ell(0, 1, 0)\right] \sim \chi^2(3)$$

  - $\ell$: the Gaussian AR(1) log-likelihood; $\hat{\mu}, \hat{\sigma}, \hat{\rho}$: its maximisers
  - $\chi^2(3)$: three restrictions; 5% critical value 7.81; a large LR rejects calibration or independence

# Prerequisites for Today (10/24): proper scoring rules

- A **score** $S(F, y)$ rates a density forecast $F$ once $y$ is observed (lower is better); it is **proper** Gneiting and Raftery (2007) if, for $Y \sim G$, no forecast beats the truth on average

$$\mathbb{E}_G\, S(G, Y) \le \mathbb{E}_G\, S(F, Y) \quad \text{for every } F$$

  - **strictly proper**: equality only for $F = G$; an improper score rewards a dishonest forecast

- **Log score**: $\mathrm{LogS}(F, y) = -\ln f(y)$, with $f$ the forecast density
  - the **Kullback–Leibler divergence** $\mathrm{KL}(g\|f) = \int g(y) \ln[g(y)/f(y)]\, dy \ge 0$ measures how far $f$ is from the true density $g$
  - $\mathrm{KL}(g\|f) = 0$ only if $f = g$

# Prerequisites for Today (11/24): the CRPS

- **CRPS** (continuous ranked probability score): the squared distance between the forecast CDF and the step at the outcome

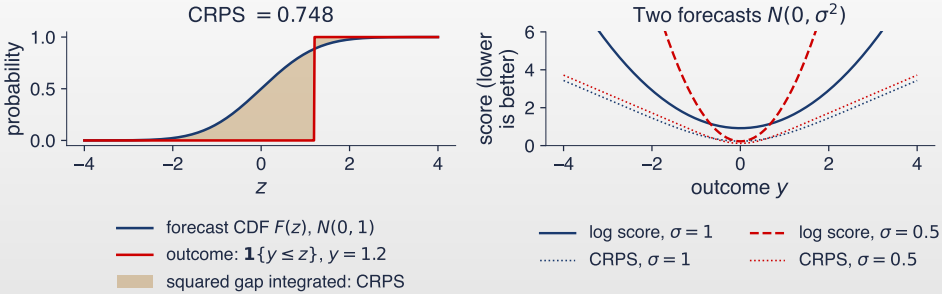
$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \big[F(z) - 1\{y \le z\}\big]^2 dz = \mathbb{E}|X - y| - \tfrac{1}{2}\mathbb{E}|X - X'|$$

  - $X, X'$: independent draws from $F$; in the units of $y$; strictly proper
- Closed form for $F = N(\mu, \sigma^2)$, with $z = (y - \mu)/\sigma$

$$\mathrm{CRPS} = \sigma\Big[z\,(2\Phi(z) - 1) + 2\varphi(z) - \frac{1}{\sqrt{\pi}}\Big]$$

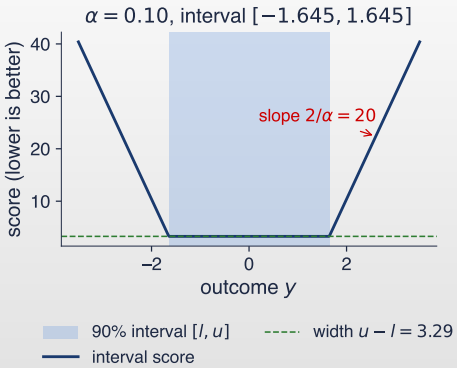
  - $\Phi$, $\varphi$: the distribution function and the density of $N(0, 1)$

# Prerequisites for Today (12/24): log score and CRPS compared




- Left: the CRPS is the area of the squared gap between $F$ and the step at $y$
- Right: far in the tails the log score grows like $z^2$, the CRPS only like $|z|$: the log score punishes overconfidence harder

# Prerequisites for Today (13/24): interval and quantile forecasts




- Central interval $[l, u]$ at level $1-\alpha$; **interval score**

$$\mathrm{IS} = (u-l) + \tfrac{2}{\alpha}(l-y)1\{y<l\} + \tfrac{2}{\alpha}(y-u)1\{y>u\}$$

  - the width plus $2/\alpha$ per unit by which $y$ falls outside
  - proper: rewards narrow intervals only if they still cover

- **Coverage**: the share of outcomes inside $[l, u]$; target $1-\alpha$
  - also report the shares below $l$ and above $u$ (each $\alpha/2$)
  - a $\tau$-quantile forecast is scored by the pinball loss $\rho_\tau$

# Prerequisites for Today (14/24): the Diebold–Mariano test

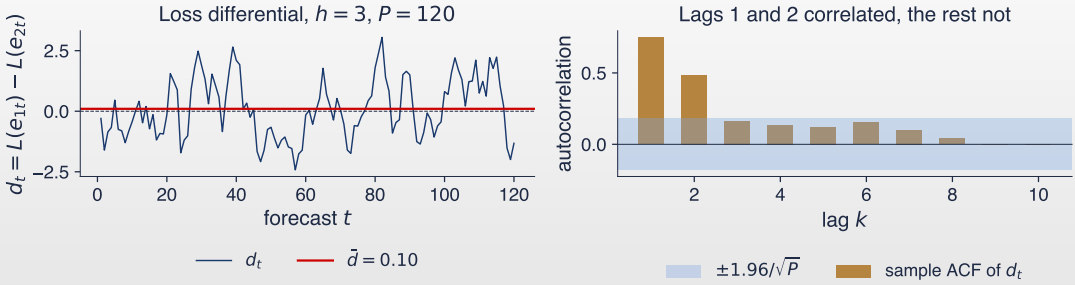
- **Diebold–Mariano** Diebold and Mariano (1995): compare two forecasts through the **loss differential**

$$d_t = L(e_{1t}) - L(e_{2t}), \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}^2_{LR}/P}}, \qquad \hat{\sigma}^2_{LR} = \hat{\gamma}_0 + 2\sum_{k=1}^{h-1} \hat{\gamma}_k$$

  - $H_0$: equal expected loss, $\mathbb{E}d_t = 0$; $\bar{d}$: the mean of $d_t$ over the $P$ forecasts
  - $\hat{\gamma}_k$: the sample autocovariances of $d_t$; $\hat{\sigma}^2_{LR}$: its long-run variance (Seminar 0) with the rectangular kernel and $h-1$ lags
  - $\mathrm{DM} > 0$: forecast 2 has the lower loss; reference distribution $N(0,1)$

- Why $h-1$ lags
  - $h$-step errors share $h-1$ future shocks with their neighbours: $d_t$ is an MA$(h-1)$
  - never use the i.i.d. variance $\hat{\gamma}_0$ for $h > 1$

# Prerequisites for Today (15/24): a loss differential at $h = 3$




- Simulated $d_t$ with expected value 0.15: overlapping three-step targets make it an MA(2)
- Right: the autocorrelations at lags 1 and 2 leave the band; they must enter $\hat{\sigma}^2_{LR}$

# Prerequisites for Today (16/24): the HLN correction and p-values

- **HLN** Harvey, Leybourne and Newbold (1997): a small-sample correction of DM

$$\mathrm{DM}^* = \mathrm{DM}\sqrt{\frac{P+1-2h+h(h-1)/P}{P}} \sim t_{P-1}$$

  - the factor removes the small-sample bias of $\hat{\sigma}^2_{LR}$
  - $t_{P-1}$: Student $t$ with $P-1$ degrees of freedom, with fatter tails than $N(0,1)$
- p-values
  - two-sided $2[1 - F_{t_{P-1}}(|\mathrm{DM}^*|)]$: are the forecasts equally accurate?
  - one-sided $1 - F_{t_{P-1}}(\mathrm{DM}^*)$: is forecast 2 better? ($F_{t_{P-1}}$: the distribution function of $t_{P-1}$)
- Several horizons or models: the Holm correction (Seminar 0)
  - $m$ tests: sort the p-values, multiply the $k$-th smallest by $m-k+1$, keep the sequence increasing

# Prerequisites for Today (17/24): nested models: Clark–West

- Model 1 (small) is **nested** in model 2 if model 2 reduces to model 1 when some coefficients are 0
  - example: the random walk inside an AR model of the changes
  - under $H_0$ the extra coefficients are 0, but the large model still estimates them, with noise
- **Clark–West** Clark and West (2007): add back the noise term to the large model

$$f_t = e_{1t}^2 - \left[e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2\right]$$

  - $\hat{y}_{it}$: the forecast of model $i$; one-sided $t$-test of $\mathbb{E}f_t = 0$ against $\mathbb{E}f_t > 0$
  - $t = \bar{f}/\mathrm{se}(\bar{f})$ compared with $N(0,1)$: reject at 5% if $t > 1.645$

# Prerequisites for Today (18/24): forecast efficiency: Mincer–Zarnowitz

- **Mincer–Zarnowitz** Mincer and Zarnowitz (1969): regress the outcome on the forecast

$$y_{t+h} = a + b\,\hat{y}_{t+h|t} + u_{t+h}$$

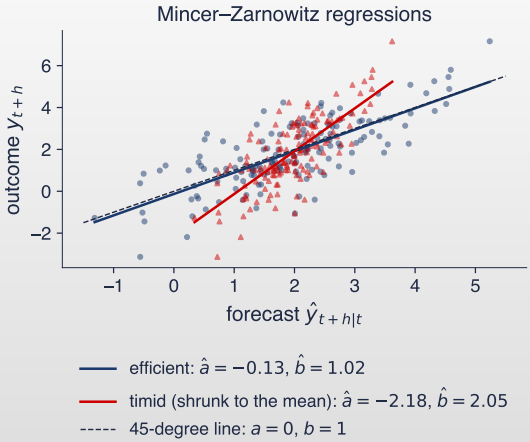
  - an efficient (optimal under squared loss) forecast has $a = 0$, $b = 1$: no systematic error left
  - $b > 1$: the forecasts move too little (too timid); $b < 1$: too much

- Joint **Wald test** of $\theta = (a, b)' = (0, 1)'$

$$W = (\hat{\theta} - \theta_{\mathbf{0}})'\,\hat{V}^{-\mathbf{1}}(\hat{\theta} - \theta_{\mathbf{0}}) \sim \chi^{\mathbf{2}}(2)$$

  - $\hat{V}$: the HAC covariance of $\hat{\theta}$ ($u_{t+h}$ is an MA($h-1$)); 5% critical value 5.99

# Prerequisites for Today (19/24): efficient and timid forecasts




- 150 simulated forecast–outcome pairs
- Blue: the efficient forecast
  - the fitted line is close to the 45-degree line
- Red: the same forecast shrunk halfway to the mean
  - $\hat{b} \approx 2$, $\hat{a} < 0$: a forecaster who moves too little
- MZ checks only the first moment: a forecast can pass it and still be a poor density forecast

# Prerequisites for Today (20/24): combining two forecasts

- Combined forecast $\hat{y}_c = w\hat{y}_1 + (1-w)\hat{y}_2$, unbiased errors with variances $\sigma_1^2$, $\sigma_2^2$, covariance $\sigma_{12} = \rho\sigma_1\sigma_2$

$$\mathrm{MSE}(w) = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}$$

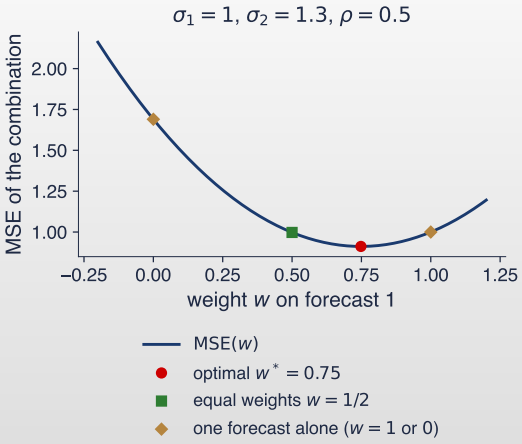
  - $\rho$: the correlation of the two forecast errors

- **Bates–Granger** Bates and Granger (1969): the weight that minimises the MSE

$$w^* = \frac{\sigma_2^2 - \sigma_{12}}{D}, \qquad \mathrm{MSE}(w^*) = \frac{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}{D}, \qquad D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$$

  - $D = \mathrm{Var}(e_1 - e_2)$; $\mathrm{MSE}(w^*) \leq \min(\sigma_1^2, \sigma_2^2)$: a worse forecast can still help
  - $w^*$ outside $[0, 1]$ is possible when the errors are strongly correlated

# Prerequisites for Today (21/24): the MSE of a combination




- $\mathrm{MSE}(w)$ is a parabola in $w$ with its minimum at $w^*$
- Near $w^*$ the curve is flat: equal weights lose very little here
- **Combination puzzle** Smith and Wallis (2009)
  - $w^*$ must be estimated from past errors; an estimated $\hat{w}$ adds about $\mathrm{Var}(\hat{w})\, D$ to the MSE
  - so equal weights often beat estimated ones

# Prerequisites for Today (22/24): combination schemes

- $N$ forecasts $\hat{y}_{1t}, \ldots, \hat{y}_{Nt}$ at each origin
  - **mean** $N^{-1} \sum_i \hat{y}_{it}$; **median**; **trimmed mean**: the mean after dropping the highest and the lowest 10%
- **Inverse-MSE weights**: better past forecasters get more weight

$$w_i = \frac{1/\mathrm{MSE}_i}{\sum_{j=1}^{N} 1/\mathrm{MSE}_j}$$

  - $\mathrm{MSE}_i$: the MSE of forecaster $i$ over a recent window of known outcomes
- **Shrinkage** towards equal weights

$$w_i^{\lambda} = \lambda \hat{w}_i + (1 - \lambda)/N, \qquad 0 \leq \lambda \leq 1$$

  - $\lambda = 0$: the mean; $\lambda = 1$: the estimated weights; **previous best**: the forecaster with the lowest past MSE Genre et al. (2013)

# Prerequisites for Today (23/24): GARCH-$t$ densities, elicitability, model sets

- **GARCH(1,1)**-$t$ Bollerslev (1987): a density forecast that follows volatility

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$$

  - $z_t$: Student $t$ with $\nu$ degrees of freedom rescaled to variance 1 (fat tails); parameters by maximum likelihood
  - benchmark: i.i.d. Normal, $r_t \sim N(\mu, \sigma^2)$ with constant $\sigma$

- **Elicitable** functional Gneiting (2011): one that is the optimal forecast under some strictly consistent loss
  - the mean (squared loss) and the quantiles (pinball) are elicitable

- **Model confidence set** (MCS) Hansen, Lunde and Nason (2011)
  - start from all models; test equal expected loss at level $\alpha$; while it is rejected, remove the significantly worst model
  - the models left form the $(1 - \alpha)$ MCS

# Prerequisites for Today (24/24): real-time data and surveys

- A **vintage**: the data as published on a given date; GDP is revised for years Croushore and Stark (2001)
  - first, second, third **release**: the first three published values of a quarter; **latest**: today's value
  - the **revision**, latest minus first release, can be large relative to the forecast error, so the choice of vintage matters
- **SPF**: the Survey of Professional Forecasters (Philadelphia Fed), quarterly since 1968
  - individual answers are public, so combinations can be built from them
  - quarterly level forecasts become an annualised growth rate: $100[(Y_{q+1}/Y_q)^4 - 1]$
- A fair evaluation scores each forecast against the vintage it was meant to predict, chosen **before** computing

# A1: optimal forecasts of an exponential variable [Solved]

## Task

- $Y \sim$ Exponential(1): $F(y) = 1 - e^{-y}$, $\mathbb{E}Y = 1$, $\mathrm{Var}Y = 1$.
- 1. Derive the optimal forecast under the squared, the absolute and the pinball loss with $\tau = 0.9$.
- 2. Give the minimum expected loss (the risk) under the squared and the absolute loss.
- 3. A forecaster reports the median and is judged by RMSE. Compute the excess expected squared loss.
- Report: three forecasts, two risks, one excess loss.

## Solution

- 1. Squared: $x^* = \mathbb{E}Y = 1$; absolute: $F(x) = 1/2 \Rightarrow x^* = \ln 2 = 0.693$; pinball: $F(x) = 0.9 \Rightarrow x^* = \ln 10 = 2.303$
- 2. Squared risk $= \mathrm{Var}Y = 1$; absolute risk $\mathbb{E}|Y - \ln 2| = \ln 2 = 0.693$ (by integration); pinball risk at $\ln 10$: 0.230
- 3. $\mathbb{E}(Y - m)^2 = \mathrm{Var}Y + (1 - m)^2$, excess $(1 - \ln 2)^2 = 0.094$: the honest median looks 9% worse under MSE
- the loss must be announced before the forecast is made

Back Exercise list

# A2: the optimal forecast under LinEx loss [Proposed]

Solution discussed in class

## Task

- The LinEx loss $L(e)$ of Prerequisites (6/24), with $e = y - x$; $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$. Model: A1.
- 1. Show that the first-order condition is $\mathbb{E}e^{a(Y-x)} = 1$ and solve it for $x$.
- 2. Compute $x^*$ for $\mu = 2$, $\sigma = 1.5$ and $a = 0.5$, then $a = -0.5$.
- 3. Explain why a rational forecaster with this loss fails the Mincer–Zarnowitz test, and why the bias moves with $\sigma_t$.
- Report: one formula, two numbers, two sentences.

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# A3: log score and CRPS of two Normal forecasts [Solved]

## Task

- Two forecasts of $y$: $N(2,1)$ and the sharper $N(2, 0.5^2)$; the outcome is $y = 3.5$.
- 1. Compute $z$, the log score $-\ln f(y)$ and the CRPS of each forecast.
- 2. Which forecast wins under each score, and which score penalises the miss more?
- 3. Prove that the log score is proper using $\mathrm{KL}(g\|f) \geq 0$.
- Report: a table of four scores, one sentence, a two-line proof.

## Solution

- 1. $N(2,1)$: $z = 1.5$, LogS $= \frac{1}{2}\ln 2\pi + z^2/2 = 2.044$, CRPS $= 0.994$
- $N(2, 0.25)$: $z = 3.0$, LogS $= 4.726$, CRPS $= 1.218$
- 2. The wider forecast wins under both; LogS more than doubles for the sharp one ($z^2$ grows fourfold), CRPS rises by about a fifth: the log score punishes overconfidence harder
- 3. $\mathbb{E}_g[-\ln f(Y)] - \mathbb{E}_g[-\ln g(Y)] = \int g \ln(g/f) = \mathrm{KL}(g\|f) \geq 0$, with equality iff $f = g$: strictly proper

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# A4: an improper score and the interval score [Proposed]

Solution discussed in class

## Task

- Truth $Y \sim N(0,1)$; forecasts $N(0, s^2)$. The *linear score* is $-f(y)$. Model: A3.
- 1. Show that $\mathbb{E}f(Y) = [2\pi(1+s^2)]^{-1/2}$ and evaluate it for $s = 0.5, 1, 2$.
- 2. Explain why the linear score is improper and what behaviour it rewards.
- 3. A 90% interval is $[1, 4]$. Compute the interval score for $y = 3$, $y = 5$ and $y = -1$.
- Report: three expectations, one sentence, three scores.

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# A5: the Diebold–Mariano test by hand [Solved]

## Task

- Two-step forecasts ($h = 2$), $P = 40$; squared-loss differential with $\bar{d} = 0.30$, $\hat{\gamma}_0 = 1.2$, $\hat{\gamma}_1 = 0.4$.
- 1. Compute the long-run variance with the rectangular kernel and $h - 1$ lags, the standard error of $\bar{d}$ and the DM statistic.
- 2. Apply the HLN correction and compare with $t_{39}$.
- 3. What would the i.i.d. variance have concluded?
- Report: two statistics, two $p$-values, one sentence.

## Solution

- 1. $\hat{\sigma}^2_{LR} = 1.2 + 2 \cdot 0.4 = 2.0$; SE $= \sqrt{2/40} = 0.224$; DM $= 0.30/0.224 = 1.342$, $p = 0.180$
- 2. factor $\sqrt{(41 - 4 + 2/40)/40} = 0.962$; HLN $= 1.291$, $p = 0.204$ ($t_{39}$ critical value 2.023)
- 3. i.i.d.: SE $= \sqrt{1.2/40} = 0.173$, DM $= 1.73$: closer to rejection, wrongly, because positive autocorrelation was ignored
- no evidence that forecast 2 is better

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# A6: Clark–West by hand [Proposed]

Solution discussed in class

## Task

- A random walk (model 1) is nested in an AR model (model 2); $P = 100$ one-step forecasts: $\overline{e_1^2} = 1.00$, $\overline{e_2^2} = 0.98$, $\overline{(\hat{y}_1 - \hat{y}_2)^2} = 0.06$; standard deviations of $f_t$ and of $d_t$: 0.5. Model: A5.
- 1. Compute the naive DM statistic for $e_1^2 - e_2^2$ and its one-sided $p$-value.
- 2. Compute the Clark–West adjusted mean $\bar{f}$, its $t$ statistic and the one-sided $p$-value.
- 3. Explain why the adjustment is added to the larger model, and why DM is undersized here.
- Report: two statistics, two $p$-values, two sentences.

# A7: Bates–Granger weights [Solved]

## Task

- Two unbiased forecasts with error standard deviations $\sigma_1 = 2$, $\sigma_2 = 3$ and error correlation $\rho = 0.4$.
- 1. Compute the optimal weight on forecast 1.
- 2. Compute the combined error variance and compare it with $\sigma_1^2$.
- 3. Compute the variance with equal weights.
- Report: one weight, two variances, one sentence.

## Solution

- 1. $\sigma_{12} = 0.4 \cdot 2 \cdot 3 = 2.4$; $D = 4 + 9 - 4.8 = 8.2$; $w^* = (9 - 2.4)/8.2 = 0.805$
- 2. $\sigma_c^2 = (36 - 5.76)/8.2 = 3.688 < 4$: the worse forecast still helps
- 3. $\frac{1}{4}(4 + 9 + 2 \cdot 2.4) = 4.45$: worse than forecast 1 alone; with very unequal forecasts equal weights can hurt

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# A8: the cost of estimated weights [Proposed]

Solution discussed in class

## Task

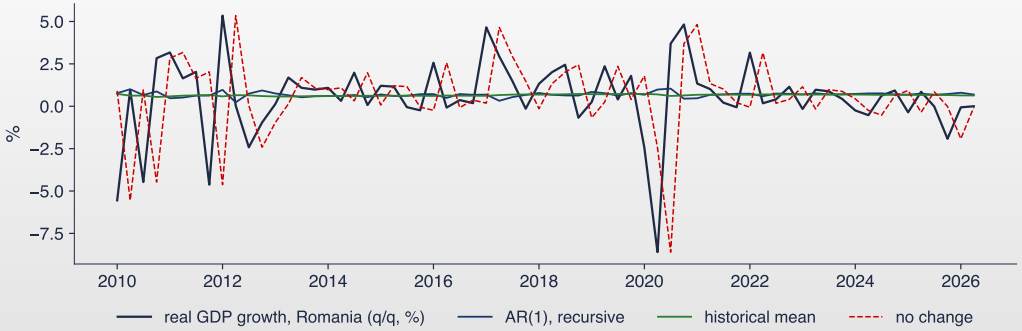
- $\sigma_1^2 = 1$, $\sigma_2^2 = 1.1$, $\rho = 0.5$; the weight is estimated from $n = 40$ past errors. Model: A7.
- 1. Show that $\mathrm{MSE}(w) = \mathrm{MSE}(w^*) + (w - w^*)^2 D$ with $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$.
- 2. Compute $w^*$, $D$ and the gain of $w^*$ over $w = 1/2$.
- 3. A simulation gives $\mathrm{Var}(\hat{w}) = 0.020$ at $n = 40$. Compare the expected estimation loss with the gain and conclude.
- Report: one identity, three numbers, one conclusion.

# B1: forecasting Romanian GDP growth [Solved]

- **Question**: does an AR(1) forecast Romanian quarterly GDP growth better than the historical mean or no change, and does the answer depend on the loss and on 2020?
- **Inputs**: real GDP growth q/q (%, Eurostat, SCA), one-step forecasts made recursively, 2010Q1–2026Q2 ($P = 66$)
- **Tasks**
  1. Compute recursive one-step forecasts from an AR(1), the historical mean and no change, and report RMSE and MAE.
  2. Test AR(1) against each benchmark with DM–HLN under squared and absolute loss.
  3. Repeat the tests without the four quarters of 2020.
  4. Interpretation: is AR(1) a better forecast than the mean, and is it better than no change?
- **Report**: a table of RMSE and MAE, eight HLN statistics with $p$-values, one sentence
- Notebook: section B1

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# B1: solution (1/2) [Solved]




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# B1: solution (2/2) [Solved]

| Forecast | RMSE | MAE | RMSE, no 2020 | MAE, no 2020 |
|---|---|---|---|---|
| AR(1) | 2.26 | 1.48 | 1.83 | 1.25 |
| mean | 2.24 | 1.48 | 1.84 | 1.26 |
| no change | 3.23 | 2.12 | 2.78 | 1.88 |

- HLN, AR(1) against the mean: 0.43 ($p$ 0.671) squared, $-0.12$ ($p$ 0.908) absolute; against no change: $-1.88$ ($p$ 0.065) and $-2.58$ ($p$ 0.012); without 2020: $-2.55$ ($p$ 0.013)
- Interpretation: AR(1) and the mean are indistinguishable (the persistence of growth is weak); both clearly beat no change, and the squared-loss evidence is weaker with 2020 because the two COVID quarters dominate the HAC variance

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# B2: combining SPF unemployment forecasts [Proposed]

- **Question**: can any combination of the individual SPF forecasts of the US unemployment rate beat the simple mean two quarters ahead?
- **Inputs**: SPF variable UNEMP4 (the quarter two quarters after the survey), FRED UNRATE quarterly average, surveys 1990–2025; model: B1 and A7
- **Tasks**
  1. Build the mean, median, 10% trimmed mean, inverse-MSE weights (last 20 surveys with a known outcome, at least 8 forecasts), their 50% shrinkage and the previous best.
  2. Report RMSE relative to the mean and DM–HLN with $h = 3$ against the mean.
  3. Run the Mincer–Zarnowitz test for the mean, with and without 2020.
  4. Interpretation: is the puzzle confirmed for unemployment, and is the mean forecast efficient?
- **Report**: six relative RMSEs, five HLN statistics, two MZ tests, one sentence
- Notebook: section B2

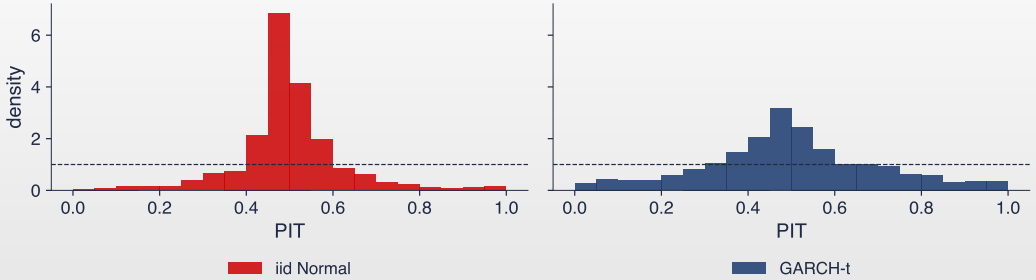
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# B3: density forecasts of EUR/RON [Solved]

- **Question**: is a GARCH(1,1)-$t$ density forecast of daily EUR/RON returns calibrated, and is it better than an i.i.d. Normal forecast?
- **Inputs**: BNR reference rate, $100\Delta \ln S_t$; estimation 2010–2018 (2 270 days), evaluation 2019–2026 (1 925 days), parameters fixed
- **Tasks**
  1. Fit both models by maximum likelihood on the estimation sample and compute one-day PITs on the evaluation sample.
  2. Plot the PIT histograms, run the Berkowitz test and report the share of PITs outside $[0.05, 0.95]$.
  3. Compute the average log score and CRPS and test their differences with DM–HLN.
  4. Interpretation: can the better-scoring model still be miscalibrated, and why?
- **Report**: two histograms, two Berkowitz tests, two shares, four scores, two HLN statistics, one sentence
- Notebook: section B3

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# B3: solution (1/2) [Solved]




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# B3: solution (2/2) [Solved]

| Model | Log score | CRPS | Berkowitz LR | $p$ | Outside 90% |
|---|---|---|---|---|---|
| i.i.d. Normal | $-0.431$ | 0.0688 | 1442 | $< 0.001$ | 1.1% |
| GARCH(1,1)-$t$ | $-1.224$ | 0.0442 | 539 | $< 0.001$ | 3.3% |

- GARCH-$t$: $\hat{\alpha} + \hat{\beta} = 0.9999$, $\hat{\nu} = 4.86$; HLN on the score differences: $-49.3$ (log score), $-56.9$ (CRPS)
- Interpretation: GARCH-$t$ wins every score by far, yet both are miscalibrated: after 2019 the BNR kept EUR/RON much calmer than in 2010–2018, so densities fixed in 2018 are too wide (a hump, few PITs in the tails); ranking and calibration answer different questions

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# B4: SPF GDP forecasts and data vintages [Proposed]

- **Question**: does the accuracy and the efficiency of the SPF next-quarter GDP forecast depend on the vintage used as the outcome?
- **Inputs**: SPF mean of $100[(\mathrm{RGDP3}/\mathrm{RGDP2})^4 - 1]$; RTDSM first, second, third release and latest vintage; targets 1990–2025 ($P = 139$); model: B1
- **Tasks**
  1. Compute the RMSE of the SPF mean against each of the four vintages.
  2. Run the Mincer–Zarnowitz test against each vintage.
  3. Recompute the RMSE against the first release and the latest vintage without 2020.
  4. Interpretation: which vintage should a pre-registered evaluation of SPF forecasts use, and why?
- **Report**: four RMSEs, four MZ tests, two RMSEs without 2020, one sentence
- Notebook: section B4

# B5: load quantiles at the evening peak [Proposed]

- **Question**: are the 90% load intervals for 19:00–20:00 calibrated, and does the expert ARX beat the weekly naive forecast in interval score?
- **Inputs**: Romanian hourly load (Energy-Charts), day-ahead forecasts, 2025–2026 ($P = 638$ days); quantiles from the last 182 errors; model: B3
- **Tasks**
  1. Compute the 5%, 50% and 95% quantile forecasts of the expert ARX and of the weekly naive at the peak hour.
  2. Report the coverage, the shares below and above, the average width and the interval score.
  3. Test the interval-score difference with DM–HLN and report the pinball loss at the median.
  4. Interpretation: is the narrower interval also the better one?
- **Report**: two coverages, four tail shares, two widths, two interval scores, one HLN statistic, one sentence
- Notebook: section B5

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# C1: a pre-registered evaluation by horizon [Proposed]

- **Question**: does a direct AR(3) beat no change for Romanian inflation at some horizon, once we correct for testing four horizons?
- **Inputs**: HICP inflation, y/y; origins from January 2013, rolling 10-year window; horizons 1, 3, 6, 12 months; model: B1
- **Tasks**
  1. Write the pre-registration: target, horizons, loss, benchmark, test, correction, in five lines, before computing anything.
  2. Compute the relative RMSE and DM–HLN for each horizon.
  3. Apply the Holm correction across the four horizons.
  4. Interpretation: what would you report if you had looked only at the best horizon?
- **Report**: a five-line plan, four relative RMSEs, eight $p$-values, a project plan
- Notebook: section C1

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# C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise forecast evaluation. The answer:
- (a) For h = 4 use Newey-West with 4 lags; the HLN correction makes the DM test more liberal in small samples.
- (b) For a point forecast (a point mass) the CRPS equals the absolute error.
- (c) Expected shortfall is elicitable because it is the mean of the tail, and means are elicitable.
- (d) To compare an AR model with the random walk it nests, use DM with Normal critical values.
- (e) A U-shaped PIT histogram means the predictive distribution is too wide.
- (f) The 90% MCS contains the best model with probability at least 90% asymptotically.
- (g) With sigma1 = 2, sigma2 = 3 and rho = 0.4 the optimal weight on forecast 1 is 0.6.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

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## Key takeaways

- The loss defines the optimal forecast: mean, median, quantile, or a biased value under LinEx
- Proper scores reward honesty; the log score punishes overconfidence hardest
- DM needs the HAC variance with $h-1$ lags and the HLN correction; nested models need Clark–West
- Estimated weights pay a price of order $1/n$: with similar forecasts, the mean wins
- An AI answer is a draft: check the lags, the direction of the PIT diagnosis, elicitability and the arithmetic

## After the seminar

- Lecture 1 develops each topic of today: loss functions and Bregman losses, calibration and proper scores, DM, GW, Clark–West, SPA and the MCS, combination and real-time data
- C1 can grow into a team project: combining Romanian inflation forecasts with a pre-registered design
- Reading: Diebold and Mariano (1995); Harvey, Leybourne and Newbold (1997); Gneiting and Raftery (2007); Smith and Wallis (2009)
- **The seminar is for practice and is not graded**

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